

RIONEGRO, Colombia

WMO: 801120

Lat: 6.165N

Lon: 75.423W

Elev: 2142

StdP: 78.10

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 11	(b) 10.2	(c) 11.1	(d) 8.0	(e) 8.6	(f) 20.1	(g) 8.9	(h) 9.2	(i) 19.3	(j) 6.8	(k) 20.2	(l) 6.1	(m) 21.0	(n) 0.2	(o) 240	(p) 0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 8.5	(c) 24.0	(d) 15.8	(e) 23.2	(f) 15.6	(g) 23.0	(h) 15.6	(i) 17.5	(j) 21.2	(k) 17.1	(l) 20.9	(m) 16.8	(n) 20.5	(o) 5.1	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.2	(b) 15.1	(c) 18.0	(d) 16.1	(e) 14.9	(f) 17.8	(g) 15.9	(h) 14.7	(i) 17.6	(j) 58.8	(k) 21.3	(l) 57.1	(m) 20.8	(n) 56.0	(o) 20.4	(p) 25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.1	(b) 6.0	(c) 5.3	(e) 7.7	(f) 26.6	(g) 2.0	(h) 1.8	(i) 6.2	(j) 28.0	(k) 5.0	(l) 29.0	(m) 3.9	(n) 30.1	(o) 2.4	(p) 31.4		
(a) 7.1	(b) 6.0	(c) 5.3	(e) 7.7	(f) 19.9	(g) 1.0	(h) 2.0	(i) 6.9	(j) 21.4	(k) 6.3	(l) 22.6	(m) 5.7	(n) 23.7	(o) 5.0	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	17.1	17.4	17.5	17.7	17.9	17.6	17.4	17.4	17.5	17.3	17.2	17.2		
	DBStd	0.97	1.04	1.11	1.05	0.95	0.95	0.85	0.81	0.85	0.83	0.97	0.91	0.96		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	358	39	29	30	24	19	24	31	29	26	34	34	37		
	CDD10.0	2715	221	208	233	230	244	228	229	231	226	226	217	223		
	CDD18.3	29	2	3	4	4	6	2	1	2	2	2	1	1		
	CDH23.3	56	4	6	7	6	9	4	3	4	6	3	1	2		
	CDH26.7	1	0	0	0	0	1	0	0	0	0	0	0	0		
(14) Wind	WSAvg	1.5	1.5	1.5	1.4	1.3	1.4	1.7	1.9	1.9	1.7	1.4	1.2	1.3		
(15) Precipitation	PrecAvg	1934	51	72	102	199	228	177	163	209	222	248	175	87		
	PrecMax	2366	109	169	227	326	376	291	284	348	324	370	348	171		
	PrecMin	1637	7	13	40	107	78	87	26	84	114	156	74	22		
	PrecStd	190	28	39	42	60	67	52	62	69	58	53	63	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	24.1	24.1	24.1	24.2	24.0	23.8	24.0	24.1	23.7	23.1	23.2		
		MCWB	15.6	15.9	16.3	16.4	16.3	15.9	15.4	15.3	15.3	15.6	15.8	15.5		
	2%	DB	22.9	23.1	23.1	23.1	23.2	23.1	23.1	23.1	23.1	22.9	22.2	22.4		
		MCWB	15.5	15.6	16.0	16.2	16.1	15.6	15.3	15.2	15.3	15.5	15.7	15.4		
	5%	DB	22.0	22.1	22.2	22.1	22.7	22.7	22.6	22.8	22.8	22.0	21.6	21.9		
		MCWB	15.3	15.5	15.9	16.2	16.1	15.5	15.1	15.2	15.2	15.4	15.7	15.4		
	10%	DB	21.0	21.2	21.2	21.2	21.8	22.0	22.0	22.0	21.9	21.0	20.8	20.9		
		MCWB	15.2	15.4	15.8	16.1	16.1	15.4	14.9	15.0	15.2	15.4	15.6	15.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.5	17.7	17.7	17.7	17.4	17.0	17.2	17.2	17.2	17.2	17.2		
		MCDB	21.3	21.2	21.2	21.1	21.7	21.2	21.4	21.5	21.2	21.1	20.7	20.7		
	2%	WB	16.7	16.8	17.0	17.2	17.1	16.7	16.2	16.4	16.6	16.5	16.6	16.6		
		MCDB	20.3	20.6	20.7	20.8	21.0	20.7	20.4	20.5	20.6	20.2	20.0	20.0		
	5%	WB	16.2	16.4	16.6	16.8	16.7	16.2	15.9	16.0	16.1	16.1	16.2	16.1		
		MCDB	19.6	19.9	20.1	20.3	20.4	20.1	20.0	20.0	20.0	19.6	19.5	19.5		
	10%	WB	15.8	16.1	16.2	16.4	16.3	16.0	15.6	15.7	15.7	15.7	15.9	15.8		
		MCDB	18.9	19.2	19.5	19.7	19.9	19.7	19.6	19.6	19.5	19.0	19.1	19.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	8.7	8.5	8.3	8.5	9.3	9.6	9.8	9.4	8.4	8.0	8.3		
		MCDBR	10.0	9.8	9.6	9.3	9.7	10.3	10.5	10.7	10.5	9.5	9.0	9.4		
		MCWBR	3.9	3.8	3.8	3.7	3.7	3.9	3.9	3.9	3.8	3.7	3.5	3.6		
	5% WB	MCDBR	8.7	8.8	8.7	8.6	8.8	9.2	9.5	9.4	9.4	8.6	8.1	8.4		
		MCWBR	3.7	3.7	3.7	3.8	3.7	3.8	3.8	3.8	3.8	3.6	3.5	3.6		
(40) Clear-Sky Solar Irradiance	taub		0.356	0.374	0.404	0.380	0.346	0.333	0.340	0.349	0.359	0.363	0.342	0.347		
	taud		2.475	2.414	2.323	2.407	2.537	2.576	2.536	2.485	2.455	2.470	2.557	2.529		
	Ebn at Noon		962	949	919	925	938	939	935	941	946	949	969	965		
	Edh at Noon		112	122	135	122	104	98	103	111	116	114	102	104		
(44) All-Sky Solar Radiation	RadAvg		4.72	4.75	4.50	4.49	4.79	5.10	5.34	5.39	5.20	4.64	4.38	4.47		
	RadStd		0.32	0.40	0.29	0.20	0.21	0.25	0.30	0.31	0.34	0.21	0.22	0.27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.24	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+0.24	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page